

Tugas Latihan Soal 2.2 (26/03/2026)

$$1. \int \frac{1}{x\sqrt{1-\ln|x|}} dx$$

misal
 $u = \ln|x|$
 $du = \frac{1}{x} dx$

$$\int \frac{1}{\sqrt{1-u^2}} du \quad \text{misal}$$

$$u = \sin \theta$$

$$du = \cos \theta d\theta$$

$$\int \frac{1}{\sqrt{1-\sin^2 \theta}} \cdot \cos \theta d\theta = \int \frac{1}{\cos \theta} \cdot \cos \theta d\theta$$

$$= \int 1 d\theta = \theta + C$$

$$u = \sin \theta \quad = \sin^{-1}(u) + C$$

$$\theta = \sin^{-1}(u) \quad = \sin^{-1}(\ln|x|) + C$$

$$2. \int e^x \sqrt{1+e^{2x}} dx$$

misal:
 $u = e^x$
 $du = e^x dx$

$$\int \sqrt{1+u^2} du \quad \text{misal } u = \tan \theta$$

$$du = \sec^2 \theta d\theta$$

$$\theta = \tan^{-1}(u)$$

$$\int \sqrt{1+\tan^2 \theta} \sec^2 \theta d\theta$$

$$\int \sec \theta \cdot \sec^2 \theta d\theta = \sec \theta \tan \theta - \int \tan \theta (\sec \theta \tan \theta) d\theta$$

$$u = \sec \theta \quad v = \tan \theta + C$$

$$= \sec \theta \tan \theta - \int \tan^2 \theta \sec \theta d\theta$$

$$= \sec \theta \tan \theta - \int (-1+\sec^2 \theta) \sec \theta d\theta$$

$$\int \sec^3 \theta d\theta = \sec \theta \tan \theta - \int \sec^3 \theta - \sec \theta d\theta$$

$$\int \sec^3 \theta d\theta = \sec \theta \tan \theta - \int \sec^3 \theta d\theta + \int \sec \theta d\theta$$

$$2 \int \sec^3 \theta d\theta = \sec \theta \tan \theta + \int \sec \theta d\theta$$

$$2 \int \sec^3 \theta d\theta = \sec \theta \tan \theta + \int \sec \theta \frac{\sec \theta + \tan \theta}{\sec \theta + \tan \theta} d\theta$$

$$2 \int \sec^3 \theta d\theta = \sec \theta \tan \theta + \int \frac{\sec^2 \theta + \sec \theta \tan \theta}{\sec \theta + \tan \theta} d\theta$$

misal
 $w = \sec \theta + \tan \theta$
 $dw = \sec \theta \tan \theta + \sec^2 \theta d\theta$

$$\int \frac{\sec^2 \theta + \sec \theta \tan \theta}{\sec \theta + \tan \theta} d\theta = \int \frac{dw}{w} = \ln|w| + C$$

$$2 \int \sec^3 \theta d\theta = \sec \theta \tan \theta + \ln|\sec \theta + \tan \theta| + C$$

$$\int \sec^3 \theta d\theta = \frac{\sec \theta \tan \theta + \ln|\sec \theta + \tan \theta|}{2} + C$$

$$\int e^x \sqrt{1+e^{2x}} dx = \frac{\sec(\tan^{-1}(u)) \tan(\tan^{-1}(u)) + \ln|\sec(\tan^{-1}(u)) + \tan(\tan^{-1}(u))|}{2} + C$$

$$= \frac{\sec(\tan^{-1}(u)) u + \ln|\sec(\tan^{-1}(u)) + u|}{2} + C$$

$$= \frac{\sec(\tan^{-1}(e^x)) e^x + \ln|\sec(\tan^{-1}(e^x)) + e^x|}{2} + C$$

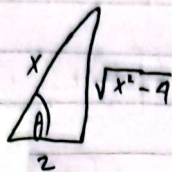
$$3. \int \frac{\sqrt{x^2 - 4}}{x} dx$$

Misal

$$x = 2 \sec \theta \rightarrow \sec \theta = \frac{x}{2}$$

$$dx = 2 \sec \theta \tan \theta d\theta$$

$$\theta = \sec^{-1}\left(\frac{x}{2}\right)$$



$$\int \frac{\sqrt{(2 \sec \theta)^2 - 2^2}}{2 \sec \theta} 2 \sec \theta \tan \theta d\theta$$

$$\int \sqrt{2^2 \sec^2 \theta - 2^2} \tan \theta d\theta$$

$$\int \tan \theta \sqrt{2^2 (\sec^2 \theta - 1)} d\theta$$

$$\int \tan \theta \sqrt{2^2 \tan^2 \theta} d\theta$$

$$\int 2 \tan^2 \theta d\theta$$

$$2 \int \tan^2 \theta d\theta$$

$$2 \int \sec^2 \theta - 1 d\theta$$

$$= 2 [\tan \theta - \theta] + C$$

$$= 2 \left[\frac{\sqrt{x^2 - 4}}{2} - \sec^{-1}\left(\frac{x}{2}\right) \right] + C$$

$$4. \int \frac{x+3}{\sqrt{x^2+3x+3}} dx$$

$$x^2+3x+3 = \left(x+\frac{3}{2}\right)^2 + \frac{3}{4}$$

$$x^2+3x+3 = x^2+3x + \underbrace{\frac{9}{4} + \frac{3}{4}}_{\frac{12}{4}=3}$$

$$\int \frac{x+3}{\sqrt{\left(x+\frac{3}{2}\right)^2 + \frac{3}{4}}} dx$$

$$\text{misalkan } u = x + \frac{3}{2} \rightarrow du = dx$$

$$x = u - \frac{3}{2}$$

$$\int \frac{u - \frac{3}{2} + 3}{\sqrt{u^2 + \frac{3}{4}}} du$$

$$\int \frac{u + \frac{3}{2}}{\sqrt{u^2 + \frac{3}{4}}} du$$

$$\int \frac{u}{\sqrt{u^2 + \frac{3}{4}}} du + \int \frac{\frac{3}{2}}{\sqrt{u^2 + \frac{3}{4}}} du$$

$$\int \frac{u}{\sqrt{u^2 + \frac{3}{4}}} du = \frac{1}{2} \int \frac{dv}{\sqrt{v}} = \frac{1}{2} \int v^{-\frac{1}{2}} dv$$

Misal

$$v = u^2 + \frac{3}{4}$$

$$dv = 2u du$$

$$\frac{1}{2} dv = u du$$

$$= \frac{1}{2} \left[\frac{v^{\frac{1}{2}}}{\frac{1}{2}} + C \right]$$

$$= \frac{1}{2} [2 \cdot v^{\frac{1}{2}} + C]$$

$$= v^{\frac{1}{2}} + C$$

$$= \sqrt{u^2 + \frac{3}{4}} + C$$

$$\sqrt{u^2 + \frac{3}{4}} + \frac{3}{2} \int \frac{1}{\sqrt{u^2 + \frac{3}{4}}} du$$

Misal

$$u = \frac{\sqrt{3}}{2} \tan \theta$$

$$du = \frac{\sqrt{3}}{2} \sec^2 \theta d\theta$$

$$\int \frac{1}{\sqrt{u^2 + \frac{3}{4}}} du = \int \frac{\frac{\sqrt{3}}{2} \sec^2 \theta d\theta}{\sqrt{\frac{3}{4} \tan^2 \theta + \frac{3}{4}}} = \int \frac{\frac{\sqrt{3}}{2} \sec^2 \theta d\theta}{\frac{\sqrt{3}}{2} \sec \theta}$$

$$u = \frac{\sqrt{3}}{2} \tan \theta$$

$$\frac{2}{\sqrt{3}} u = \tan \theta$$



$$= \int \sec \theta d\theta$$

$$= \ln |\sec \theta + \tan \theta| + C$$

$$= \ln \left| \sqrt{\frac{4u^2+3}{3}} + \frac{2}{\sqrt{3}} u \right| + C$$

$$= \sqrt{u^2 + \frac{3}{4}} + \frac{3}{2} \ln \left| \sqrt{\frac{4u^2+3}{3}} + \frac{2}{\sqrt{3}} u \right| + C$$

$$= \sqrt{\left(x+\frac{3}{2}\right)^2 + \frac{3}{4}} + \frac{3}{2} \ln \left| \sqrt{\frac{4\left(x+\frac{3}{2}\right)^2+3}{3}} + \frac{2}{\sqrt{3}} \left(x+\frac{3}{2}\right) \right| + C$$

$$5. \int \sqrt{3-2x-x^2} dx$$

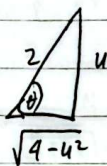
$$\begin{aligned} 3-2x-x^2 &= -x^2-2x+3 \\ &= -x^2-2x-1+4 \\ &= -(x+1)^2+4 \\ &= 4-(x+1)^2 \end{aligned}$$

$$\int \sqrt{4-(x+1)^2} dx$$

$$\begin{aligned} u &= x+1 \\ du &= dx \end{aligned}$$

$$\int \sqrt{4-u^2} du$$

$$\begin{aligned} u &= 2 \sin \theta \rightarrow \frac{u}{2} = \sin \theta \\ du &= 2 \cos \theta d\theta \quad \theta = \sin^{-1}\left(\frac{u}{2}\right) \end{aligned}$$



$$\int \sqrt{4-4\sin^2\theta} \cdot 2 \cos \theta d\theta$$

$$\int \sqrt{2^2 \cos^2\theta} \cdot 2 \cos \theta d\theta$$

$$\int 2^2 \cos^2\theta d\theta$$

$$4 \int \frac{1+\cos 2\theta}{2} d\theta$$

$$2 \left[\frac{\theta + \sin \theta \cos \theta}{2} \right] + C$$

$$= 2\theta + 2 \sin \theta \cos \theta + C$$

$$= 2 \sin^{-1}\left(\frac{u}{2}\right) + u \cdot \frac{\sqrt{4-u^2}}{2} + C$$

$$= 2 \sin^{-1}\left(\frac{x+1}{2}\right) + \frac{(x+1)\sqrt{4-(x+1)^2}}{2} + C$$

$$6. \int \frac{1}{\sqrt{x}-\sqrt[3]{x}} dx \quad \text{misal } u = x^{\frac{1}{6}}$$

$$u^6 = x$$

$$\int \frac{6u^5 du}{u^{\frac{1}{2}} - u^{\frac{1}{3}}} = \int \frac{6u^5 du}{u^{\frac{3}{6}} - u^{\frac{2}{6}}} = \int \frac{6u^5 du}{u^{\frac{1}{2}} - u^{\frac{1}{3}}}$$

$$\int \frac{6u^5 du}{u^{\frac{1}{2}} - u^{\frac{1}{3}}} = \int \frac{6u^5 du}{u^{\frac{1}{6}}(u^{\frac{1}{2}-\frac{1}{3}} - 1)} = \int \frac{6u^5 du}{u^{\frac{1}{6}}(u^{\frac{1}{6}} - 1)}$$

$$6 \int u^2 + u + 1 + \frac{1}{u-1} du$$

$$6 \left[\frac{u^3}{3} + \frac{u^2}{2} + u + \ln|u-1| \right] + C$$

$$= 6 \left[\frac{x^{\frac{3}{6}}}{3} + \frac{x^{\frac{2}{6}}}{2} + x^{\frac{1}{6}} + \ln|x^{\frac{1}{6}}-1| \right] + C$$

$$= 2\sqrt{x} + 3x^{\frac{1}{3}} + 6x^{\frac{1}{6}} + 6\ln|x^{\frac{1}{6}}-1| + C$$

$$7. \int_0^{\ln 2} \sqrt{e^x-1} dx$$

$$u = \sqrt{e^x-1} \quad u^2 = e^x-1$$

$$x=0 \Rightarrow u = \sqrt{e^0-1} = 0 \quad u^2+1 = e^x$$

$$x=\ln 2 \Rightarrow u = \sqrt{e^{\ln 2}-1} = 1 \quad 2u du = e^x dx$$

$$2u du = u^2+1 dx$$

$$\frac{2u}{u^2+1} du = dx$$

$$\int_0^1 u \cdot \frac{2u}{u^2+1} du \quad u^2+1 \sqrt{\frac{u^2}{u^2+1}} = 1$$

$$\int_0^1 \frac{2u^2}{u^2+1} du$$

$$2 \int_0^1 \left(1 - \frac{1}{u^2+1} \right) du$$

$$2 \left[u - \tan^{-1}(u) \right]_0^1$$

$$2 \left[(1 - \tan^{-1}(1)) - (0 - 0) \right] = 2 \left[1 - \frac{\pi}{4} \right] = 2 - \frac{\pi}{2}$$